

Experiment 'p_13_6_recursive_find' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_13_6_recursive_find

Natural language statement: Write a recursive method that tests whether a given text contains a string. For example, find("Mississippi", "sip") returns true. Hint: If the text starts with the string you want to match, then you are done. If not, consider the text that you obtain by removing the first character.

Method signature: p_13_6_recursive_find(text: string, str: string) returns (found: bool)

Ensures

- found == find(text, str)

Requires

- 1 <= |str| <= |text|

Functional Code Given

```
function find(text: string, str: string): bool{
  if |text| < |str| then false
  else if text[0 .. |str|] == str then true
  else find(text[1..], str)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a recursive method that tests whether a given text contains a string. For example,
→ find("Mississippi", "sip") returns true. Hint: If the text starts with the string
→ you want to match, then you are done. If not, consider the text that you obtain by
→ removing the first character.

The signature should be:

p_13_6_recursive_find(text: string, str: string) returns (found: bool)

The method should respect the following contract:

requires 1 <= |str| <= |text|, ensures found == find(text, str)

The contract uses the following dafny code:

```
function find(text: string, str: string): bool{
  if |text| < |str| then false
  else if text[0 .. |str|] == str then true
  else find(text[1..], str)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnSIZfPRyKRDbCHiS7m8bcLoUHMWJ

Model: gpt-5.5-2026-04-23

Created at: 1780679223

Finish reason: stop

Usage: 532 tokens in, and 1106 tokens out

```
{
  if text[0 .. |str|] == str {
    found := true;
  } else if |text| == |str| {
    found := false;
  } else {
    found := p_13_6_recursive_find(text[1..], str);
  }
}
```

```
}
```

Final Program

```
function find(text: string, str: string): bool{
  if |text| < |str| then false
  else if text[0 .. |str|] == str then true
  else find(text[1..], str)
}

method p_13_6_recursive_find(text: string, str: string) returns (found: bool)
  requires 1 <= |str| <= |text|
  ensures found == find(text, str)
{
  if text[0 .. |str|] == str {
    found := true;
  } else if |text| == |str| {
    found := false;
  } else {
    found := p_13_6_recursive_find(text[1..], str);
  }
}
```

Total Token Usage

Input tokens: 532

Output tokens: 1106

Reasoning tokens: 1015

Sum of 'total tokens': 1638

Experiment Timings

Overall Experiment started at 1780679212375, ended at 1780679281418, lasting 69043ms (69.04 seconds)

Iteration #1 started at 1780679212379, ended at 1780679281412, lasting 69033ms (69.03 seconds)